

Data-Aided Channel Estimation for Robust DECT-2020 NR+ Links in Oil and Gas Equipment Data Communication

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Abstract—Oil and gas extraction and processing environments impose stringent requirements on wireless data links used for equipment monitoring, control, and safety. Multipath-rich metallic structures, Radio Frequency(RF) blocking, and dynamic interference degrade channel estimation accuracy, especially at low SNR. This paper adapts and evaluates a data-aided least-squares (LS) channel estimation approach—originally developed and validated for Digital Enhancement Cordless Telecommunication-2020 New Radio Plus(DECT-2020 NR+)—to oil and gas equipment data communication over Orthogonal Frequency Division Multiplexing(OFDM) links. By combining pilot symbols with previously decoded data symbols, the proposed receiver improves channel transfer function (CTF) estimates without increasing pilot overhead. MATLAB simulations on AWGN and 3GPP TDL channels, coupled with a lab validation using a

DECT development board and a backscatter-like test signal, show consistent MSE gains over pilot-only LS across SNRs. Deployment aspects for oil and gas sites, including subcarrier numerology, packetization, and frequency planning in the 1.9 GHz frequency band, which is used in some industrial wireless systems. Detailed implementation guidance is provided for field gateways and sensor Radio Devices(RDs). Results indicate that data-aided estimation enhances reliability and spectral efficiency for real-time equipment telemetry and alarm reporting in harsh oilfield environments.

Index Terms—DECT-2020 NR+, Oil and Gas Equipment, OFDM, Channel Estimation, Data-aided, Decision-Directed Channel Estimation, LS, Industrial IoT

I. INTRODUCTION

Wireless data communication in oil and gas facilities supports distributed sensing (pressure, temperature, vibration), condition-based maintenance, mobile workforce tools, and safety interlocks. Metallic infrastructures (pipelines, tanks, derricks) and rotating machinery create severe multipath, shadowing, and non-stationary fading that stress channel estimation (CE), particularly for compact, battery-powered sensor nodes operating at low Equivalent Isotropically Radiated Power(EIRP). DECT-2020 NR+—a non-cellular 5G technology standardized by European Telecommunications Standards Institute(ETSI)—offers licensed-exempt operation around 1.9 GHz, OFDM waveforms, mesh networking, and deterministic low-latency options suitable for plant-scale industrial IoT [1].

Traditional pilot-assisted CE (e.g., LS, MMSE) suffers in low SNR or sparse-pilot regimes common in DECT NR+ profiles [2]. Data-aided or Decision-Directed Channel Estimation(DDCE) augments pilots with reliable data symbol decisions to refine the Channel Transfer Function(CTF) estimate over time, reducing mean squared error (MSE) and improving equalization [3]. Building on our thesis work in DECT NR+, this paper contributes:

- A oil and gas centric adaptation of data-aided LS CE for DECT-2020 NR+ OFDM links, preserving low complexity and pilot overhead.
- A simulation study on AWGN and Tapped Delay Line(TDL)-A/C channels across SNRs, showing MSE gains vs. pilot-only LS.
- Deployment guidance for oil and gas equipment communication, including numerology choices, packet design, and mesh routing implications.

II. BACKGROUND AND RELATED WORK

A. DECT-2020 NR+ for Industrial Sites

DECT-2020 NR+ employs CP-OFDM with configurable subcarrier spacing (27–216 kHz), TDD frames (10 ms, 24 slots), and flexible PCC/PDC mapping [4]. It enables mesh topology among radio devices (RDs) that can act as Fixed Terminals(FT) or Portable Terminals(PT), allowing self-organizing coverage within plants. Compared to cellular, DECT NR+ reduces infrastructure dependence and simplifies local control, advantageous for private oilfield networks.

B. Channel Estimation in OFDM

Channel Estimation (CE) is the technique which could get the Channel State Information (CSI) in a wireless communication channel. In the experiment in real life, the CSI cannot be obtained accurately, hence the procedure to get the factor of CSI is called Channel Estimation.

The DECT-2020 system takes advantages of OFDM symbols for communication. It is the OFDM-based system. OFDM is a multiplexing scheme that splits the data stream into multiple narrowband channels to effectively take advantages of the available bandwidth.

LS estimation divides received pilots by known pilots, offering low complexity but noise sensitivity. MMSE incorporates

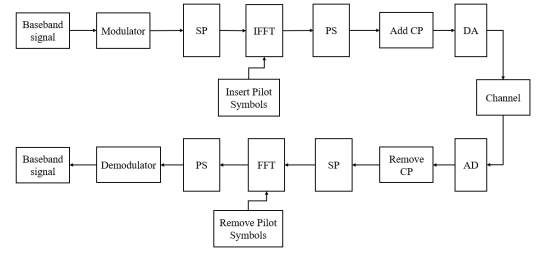


Fig. 1. OFDM system with Channel Estimation

channel/noise statistics for lower MSE but at higher complexity and with prior knowledge requirements. Data-aided CE iteratively refines estimates using detected data symbols, mitigating sparse-pilot limitations and improving low-SNR robustness.

C. Oil and Gas Environment Challenges

Oilfield and refinery layouts exhibit:

- High-RCS metallic clutter driving frequency-selective fading.
- Time variation from rotating equipment and moving vehicles.
- RF coexistence with trunked radios and Wi-Fi (2.4/5 GHz), favoring 1.9 GHz for reduced congestion.

These factors motivate CE methods that maintain accuracy under pilot scarcity and SNR variability.

III. TECHNICAL DETAILS

A. System Assumptions and Numerology

This paper target DECT-2020 NR+ with $\mu=1$ (27 kHz subcarrier spacing), FFT size $N_{\text{FFT}}=64$, CP length $N_{\text{CP}}=8$, and occupied subcarriers $N_{\text{OCC}}=56$. Frames contain PCC (control) and PDC (payload), with DRS-like pilots interleaved per ETSI mapping [5]. Modulation uses QPSK for robustness; higher Modulation and Coding Scheme(MCS) can be enabled where SNR allows. The working frequency is 1.88–1.90 GHz (regional regulations apply).

B. Channel and Signal Model

For subcarrier k and OFDM symbol l :

$$Y_k^l = H_k^l X_k^l + V_k^l \quad (1)$$

where H_k^l is the Channel Transfer Function, X_k^l is the pilot or data symbol, and V_k^l AWGN plus residual interference. Equalization uses zero-forcing (ZF):

$$X_k^l = \frac{Y_k^l}{H_k^l} \quad (2)$$

The resource grid and indexing could be seen as follows:

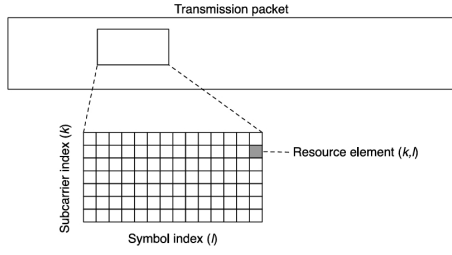


Fig. 2. Resource grid and indexing [6]

C. Pilot-Only LS Estimation

LS algorithm is the easiest and most straightforward method of channel estimation in OFDM system. It assesses the channel response by minimizing the square error between the received pilot signal and the expected known pilot signal. Due to its low complexity and low demand for channel prior statistics, LS algorithm is widely utilized in channel estimation. The result of LS estimation could be used in ZF equalization.

The equation of the LS estimation could be seen as follows [7]:

$$\hat{H}_{P,LS} = X_P^{-1} Y_P \quad (3)$$

The $\hat{H}_{P,LS}$ is the channel response of LS method at the pilot subcarriers. X_P is the known transmitted pilot symbols. While Y_P is the real pilot symbols passing the channel. The expanded matrix form could be seen as follows,

$$\hat{H}_{P,LS} = \begin{bmatrix} \frac{Y_P(1)}{X_P(1)}, & \frac{Y_P(2)}{X_P(2)}, & \dots, & \frac{Y_P(N_P)}{X_P(N_P)} \end{bmatrix}^T \quad (4)$$

Where the N_P is the number of pilot subcarriers. It means that each pilot subcarrier channel is estimated by dividing the received pilot symbols by known transmitted pilot symbols [8]. This is low-complexity but noise-sensitive, especially with sparse pilots [9].

D. Data-aided Channel Estimation

Compared with pilot-only channel estimation, the data-aided channel estimation utilize not only the pilot signal, but also the whole data packet to evaluate channel response. After decoding the data, the obtained data will be reencoded to perform data-aided channel estimation. The encoding method is same as at the transmitter side. After the signal mapping into the OFDM symbols, the channel estimation could be obtained as:

$$H = \frac{Y_{re}}{X_{en}} \quad (5)$$

The Y_{re} is the received symbols and X_{en} is the encoded data.

E. Receiver Enhancements in oil and gas filed environments

In oil and gas environments characterized by strong metallic multipath, slow time variation, and impulsive Electromagnetic Interference(EMI), the receiver is hardened by integrating environment-aware tactics atop data-aided LS: the reliability gate τ is adapted from a coarse SNR estimate derived via PCC RSSI and noise variance so that τ is stricter at low SNR to

prevent error propagation and looser at high SNR to harvest more virtual pilots (combined with CRC gating to ensure only validated codeblocks contribute); outliers are suppressed by applying subcarrier-wise median or gentle Savitzky–Golay filtering to the estimated CTF trajectories (with phase unwrapping) to remove impulsive spikes from motors and igniters without blurring genuine fades; unreliable resources are excluded by dropping 1–2 edge subcarriers per side and masking bins flagged by a short-window periodogram as narrowband interferers so they are neither used in the reliable set nor in interpolation, with persistence of masks informing agile frequency reselection; and mesh-aware cooperation is enabled in DECT-2020 NR+ by feeding per-link CE quality (CTF MSE, BER) to the controller to trigger FT/PT role switching, rerouting, or channel reselection around shadowed equipment, while tuning τ and temporal smoothing α (e.g., 0.8–0.9 on weak hops) to stabilize estimates—altogether delivering more accurate, interference-robust channel estimates and lower BLER at edge SNRs without increasing pilot density or computational burden.

IV. RESULTS

A. Simulation Setup

Under the simulation setup, oil and gas site wireless conditions are emulated by combining a flat AWGN baseline with 3GPP TDL-A/C fading at 1.9GHz, setting delay spreads between roughly 30 and 300ns and low Doppler in the 1–30Hz range to reflect slow machinery motion and vehicular activity; the SNR is swept from 0 to 30dB in 3dB steps. The PHY follows DECT-2020 NR+ $\mu=1$ numerology (27kHz spacing, $N_{FFT}=64$, $N_{CP}=8$, 56 occupied subcarriers), with ETSI pilot mapping, PCC using a Type-2 control field and turbo coding, and PDC using QPSK with turbo coding plus CRC24A, all within the standard 10 ms TDD frame. For each SNR point, at least 1000 packets are transmitted to ensure stable statistics, and results are averaged across subcarriers and OFDM symbols. The receiver first forms an initial pilot-only LS estimate $\hat{H}(0)$, applies ZF equalization and soft decoding, then performs data-aided LS by augmenting pilots with reliability-gated decoded data (LLR/CRC-based) to obtain an updated $\hat{H}(1)$, with optional subcarrier-wise median filtering, narrowband interferer masking, edge-subcarrier exclusion, and temporal exponential smoothing to stabilize slow fading. Performance is assessed by channel-transfer-function MSE against the simulator's ground-truth H , along with PDC BLER/BER to verify that decision feedback does not induce error floors, and computational effort is accounted for via operation counts to demonstrate that the proposed method remains near-LS in complexity and suitable for embedded industrial RDs.

B. Simulation Results

The channel response for 3 packets is first simulated in 3D as shown below. In the simulation, the true channel response is set to 1, and the SNR is set to 21 dB. As observed, the estimated channel response fluctuates around 1 due to the

added Gaussian white noise. If the SNR were set to infinity, the channel response would be a flat surface always equal to 1.

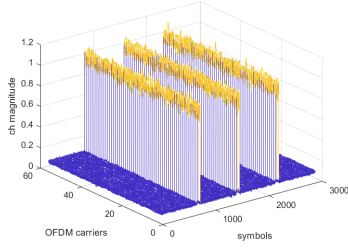


Fig. 3. Channel Response in simulation

For TDL-A channel, the channel response for one packet could be seen as follows, different from the flat AWGN channel, the surface fluctuates along the OFDM carrier axis, which indicates that different subcarriers experience different gains.

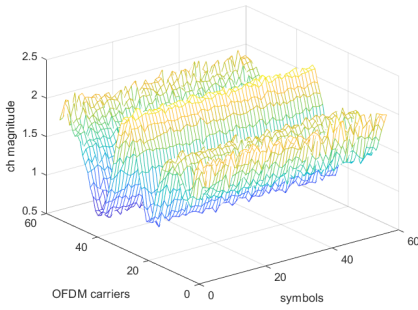


Fig. 4. Channel Response of Data-aided CE in 3D grid in simulation

Compared with Data-aided channel estimation, the Pilot-only channel estimation is not smooth enough as follows,

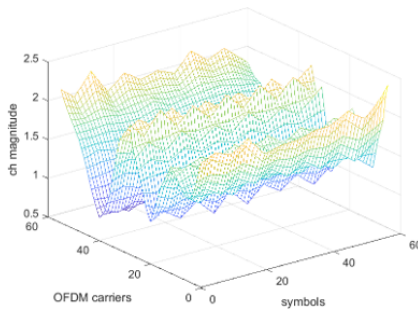


Fig. 5. Channel Response of Pilot-only CE in 3D grid in simulation

To compare the performance between pilot-only channel estimation and data-aided channel estimation. The Mean Square Error (MSE) is used to evaluate the performance of different methods of channel estimation [10]. It quantifies how accu-

rately the estimated channel response approximates the true channel response. The calculation of MSE could be seen as:

$$MSE = \frac{1}{N} \sum_{i=1}^N \|H_i - \hat{H}_i\|^2 \quad (6)$$

The $\hat{H}_i$ is the estimated channel, H_i is the true channel response. As the channel information is known, the H_i could be obtained by the function `ChannelResponseOutput` in matlab. For data aided channel estimation, one sample is an OFDM symbol, N is the number of subcarriers in OFDM. The pilot-only method only estimates channel at pilot position, the length of pilot-only method is not the same as data-aided one, the vector of pilot-only estimation in one packet needed to be inserted linear values, and transfer into the shape of OFDM grid, one sample is also the vector of one OFDM symbol.

In the simulation, the transfer of 1000 packets is imitated, including 58000 samples, each sample is an OFDM symbols, the result could be obtained by calculate the average of the squared errors across all 58000 samples. After calculating the MSE of each sample for both channel estimation methods, the average value of all samples could be obtained to evaluate the performance of two methods.

The SNR values of channel are set from 0 to 30, the interval is 3 dB, to demonstrate the MSE for both methods in different channel environment, the results could be seen as Fig.6.

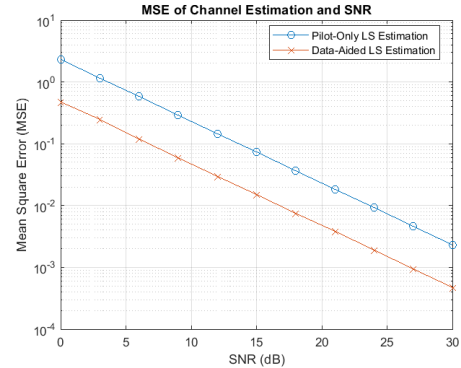


Fig. 6. MSE of Channel Estimation for AWGN channel

The results clearly show that the data-assisted LS estimation has a lower MSE than the pilot-only estimation across all SNR values. This suggests that incorporating data symbols into the estimation process significantly improves the accuracy of the channel estimation. Both methods show an exponential decrease in MSE with increasing SNR, which is reflected in the almost linear trend on the logarithmic scale. This is in line with theoretical expectations, as a higher SNR leads to a more accurate estimation. The performance difference between the two methods is particularly pronounced at low SNR values, where the pilot-only LS estimation struggles with higher errors, further highlighting the advantage of using data symbols in channel estimation. Overall, the figure highlights the superior performance of data-based estimation, which is

a more effective method for reducing estimation errors in practical communication systems.

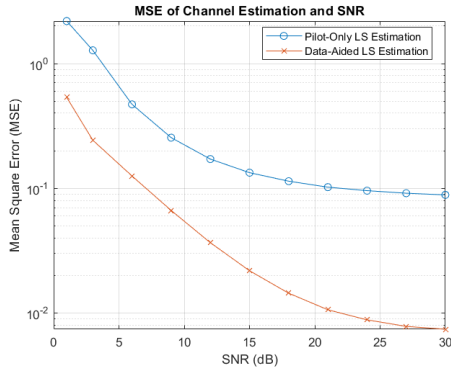


Fig. 7. MSE of Channel Estimation for nrTDL channel

For nrTDL channel, similar result of MSE could be obtained, the performance of data-aided channel estimation is better than pilot-only channel estimation for all SNR environment. The difference is that the line does not show an linear trend in this channel.

V. CONCLUSION

A data-aided LS channel estimation scheme is presented for DECT-2020 NR+ tailored to oil and gas equipment data communication. By augmenting pilot-based LS with reliability-gated decoded data, the receiver achieves lower CTF MSE and improved decoding performance under low SNR and frequency-selective fading, without increasing pilot overhead or computational burden. Simulations on AWGN and TDL channels confirm consistent gains and practical deployability in metallic, multipath-rich oil and gas environments. Future work includes fusion with adaptive MMSE/Kalman filters, MIMO extensions for spatial diversity around large equipment, and field trials in live refinery networks to quantify end-to-end reliability and energy benefits. In oil and gas operations, these improvements directly support more reliable telemetry for safety interlocks and condition-based maintenance, even amid rotating machinery and heavy steel infrastructure. The approach also aligns with the sector's power and spectrum constraints, enabling longer sensor lifetimes and denser mesh deployments without sacrificing link robustness.

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